Uncertainty Visualisation for Signal Supression

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Abstract

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This thesis is an original work of my research and contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

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Publications during enrolment

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The material in Chapter 1 has been submitted to the journal *Journal of Impossible Results* for possible publication.

The contribution in Chapter 2 of this thesis was presented in the International Symposium on Nonsense held in Dublin, Ireland, in July 2022.

Reproducibility statement

This thesis is written using Quarto with renv (Ushey 2022) to create a reproducible environment. All materials (including the data sets and source files) required to reproduce this document can be found at the Github repository `github.com/SusanSu/thesis`.

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This thesis includes ?? original papers published in peer reviewed journals and ?? submitted publications. The core theme of the thesis is ??. The ideas, development and writing up of all the papers in the thesis were the principal responsibility of myself, the student, working within the Department of Econometrics & Business Statistics under the supervision of ??

(The inclusion of co-authors reflects the fact that the work came from active collaboration between researchers and acknowledges input into team-based research.)

In the case of (??insert chapter numbers) my contribution to the work involved the following:

Thesis chapter	Publication title	Status	Nature and % of student contribution	Nature and % of coauthors' contribution	Coauthors are Monash students
2	The life cycle of Mongolian crickets	Submitted	Concept and data analysis, writing first draft: 60%	Shu Xu, input into manuscript: 25%; Eddie Betts, input into manuscript: 15%	Shu Xu: No; Eddie Betts: Yes

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Acknowledgements

I would like to thank my pet goldfish for . . .

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You may also wish to acknowledge significant and substantial contribution made by others to the research, work and writing represented and/or reported in the thesis. These could include significant contributions to: the conception and design of the project; non-routine technical work; analysis and interpretation of research data; drafting significant parts of the work or critically revising it to contribute to the interpretation.

Chapter 1

Introduction

- overview of thesis
- what is this thesis about (what problem is it solving)
- how do the chapters fit together
- Uncertainty visualisation is important
  - Decision making
  - correctly hedge conclusions drawn from graphics
- Uncertainty visualisation is a relatively new field, but it has been poorly defined
- Uncertainty itself is a vague term, each paper seems to take the meaning of it as a given, so the visualisation of it is fraught with disagreement
- Two distinct camps have established themselves in the literature. The first view uncertainty visualisation as a visualisation of an error, probability, or distribution. They treat uncertainty as a variable of interest, distinct from its estimate. Examples of this approach include ___.
- The second camp sees uncertainty as noise. They believe uncertainties role in visualisation is to dampen misleading or false conclusions, and amplify correct conclusions. They treat uncertainty as a necessary nuisance that should be communicated alongside it's estimate. Examples of this approach include ___.
- These distinct camps create several gaps in the literature
- Uncertainty visualisation are often bespoke and are not properly integrated into the grammar of graphics. The elusive definition of uncertainty prevents it from being integrated into the grammar of grpahics, which means its role in a visualisation is never clear.

- This leads to visualisation approaches that jump randomly between viewing uncertainty as a variable in or itself, or as a necessary nuisance required to property visualise an estimate. This leads software that visualise uncertainty behaving inconsistently when deciding how to display uncertainty in the graphic. Authors of the software are often unaware of the distinction between the two methods.
- The conflicting motivations also make it difficult to evaluate uncertainty visualisations
- In graphics research, user studies are king and allow us to understand which graphics convey information most effectively
- Uncertainty visualisation presents a unique issue. Most research tries to evaluate users ability to effectively extract information from a plot to establish a hierarchy of visualisation techniques
- Uncertainty visualisation requires signal amplification when uncertainty is low, and obfuscation when uncertainty is high. This means that the best visualisation technique is not one that allows for the most accurate statistic extraction.
- Some papers have specifically tried to evaluate uncertainty, but without a clear definition of uncertainty, this have proven to be difficult.
- This thesis attempts to close these gaps by providing a comprehensive analysis of uncertainty visualisation, including their definition, practical application, and evaluation.
  - First we comb the literature and discuss the motivations behind uncertainty visualisation, and identify motivations for the field.
  - Next, we formalise uncertainty visualisation, by identifying the two fields, discussing their difference and which is more appropriate to be considered “uncertainty visualisation”
  - Then, we formalise the second camp and coin the term “signal supression” to describe visualisation approaches that fall within it.
  - Then we discuss uncertainty’s place in the grammar of graphics, and describe what it means for a visualisation to be an “uncertainty visualisation” within this framework.
  - Then we introduce software, ggdibbler, which is the practical application of the theoretical inclusion of uncertainty in the grammar of graphics
  - Finally, we formalise an approach for evaluating the effectivness of uncertainty visualisation, and use this tool to evaluate existing approaches to uncertainty visualisation.

1.1 Thesis Outline

The thesis is structured as follows.

sec-first-paper provides a comprehensive review of the uncertainty visualsiation literature. Existing reviews in the field take the purpose of an uncertainty visualisation to be self evident which results in a large amount of conflicting information. The most common approach communicates uncertainty as a probability or a distribution, and focus on showing the uncertainty as an isolated variable disconnected from its context. We contrast these visualisations with approaches that view uncertainty as noise that should be incorporated to give a holistic view of our data. We coin the term "signal suppression" to describe a visualisation that is designed for preventing false conclusions, as the approach demands that the signal (i.e. the conclusions drawn from the estimates) is suppressed by the noise (i.e. the variance on those estimates). We provide motivation for viewing signal supression as the most worthwhile goal in uncertainty visualisation, and argue that visualisations that display uncertainty as an isolated variable should not be considered uncertainty visualisations at all. We further discuss difficulties in creating and evaluating the effectiveness of these plots, which further motivates the following chapters of this thesis.

sec-second-paper introduces a new R package, ggdibbler. In sec-first-paper we discussed the run on effects of the ambiguous definition of uncertainty. These problems included difficulty incorperating uncertainty into the grammar of graphics framework, which leads to software that does not always behave in ways that users would expect. sec-second-paper closes this gap by discussing the theoretical implications of including uncertainty in the grammar of graphics, and introducing ggdibbler, a ggplot extension that allows us to visualise uncertainty holistically and prevent false signals. We illustrate how ggdibbler can be seamlessly integrated into existing visualisation workflows and demonstrate its usefulness by looking at examples that use ggdibbler to incorporate uncertainty into a choropleth map.

sec-third-paper presents a user study, motivated by problems faced by the Australian Energy Market Operator (AEMO), that evaluates the effectiveness of different uncertainty visualisations. sec-first-paper highlighted the difficulty in evaluating uncertainty visualisations, as the uncertainty must be evaluated as noise rather than signal, and provides some suggestions for experimental methods. This section applies the suggested experimental methods and identifies which graphics, implemented by the ggdibbler software introduced in sec-second-paper, are effective tools for signal suppression. The experiment in this section is motivated by a case study specific to problems faced by AEMO. The transition to renewable energy results in a large amount of uncertainty in Australia's energy demands,

which also demands better understanding of the uncertainty if we are to make the transition without issue.

sec-first-paper is an invited contribution to Volume 14 of *Annual Reviews of Statistics and it's Applications*. The ggdibbler software introduced in sec-second-paper has been submitted to CRAN, and the associated paper will be submitted to the Journal of Computational and Graphical Statistics upon it's completion.

Chapter 2

Literature Review

This chapter contains a summary of the context in which your research is set.

Imagine you are writing for your fellow PhD students. Topics that are well-known to them do not have to be included here. But things that they may not know about should be included.

Resist the temptation to discuss everything you've read in the last few years. And you are not writing a textbook either. This chapter is meant to provide the background necessary to understand the material in subsequent chapters. Stick to that.

You will need to organize the literature review around themes, and within each theme provide a story explaining the development of ideas to date. In each theme, you should get to the point where your ideas will fit in. But leave your ideas to later chapters. This way it is clear what has been done beforehand, and what new contributions you are making to the research field.

All citations should be done using markdown notation as shown below. This way, your bibliography will be compiled automatically and correctly.

2.1 Exponential smoothing

Exponential smoothing methods were originally developed in the late 1950s (Brown 1959, 1963; Holt 1957; Winters 1960). Because of their computational simplicity and interpretability, they became widely used in practice.

Empirical studies by Makridakis & Hibon (1979) and Makridakis et al. (1982) found little difference in forecast accuracy between exponential smoothing and ARIMA models. This made the family of exponential smoothing procedures an attractive proposition (see Chatfield et al. 2001).

The methods were less popular in academic circles until Ord, Koehler & Snyder (1997) introduced a state space formulation of some of the methods, which was extended in Hyndman et al. (2002) to cover the full range of exponential smoothing methods.

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